

About This Certification

The United Endurance Sports Coaching Academy's (UESCA) running certification is a comprehensive running certification that will provide you with everything you need to know to be an effective and professional running coach. As a certified running coach, your client will expect you to know up-to-date information on everything from what type of shoes to recommend for a particular event to how to integrate yoga into a running program. This certification teaches all these things and more. After obtaining this certification, you will be a comprehensive resource for a client who wants to train for running ... even if you have no previous knowledge of the sport.

As the sport of running grows in popularity, so does the demand for qualified coaches. This certification will prepare you to answer almost any question or concern your clients may have.

The goal of this certification is to create a well-rounded running coach who understands the fundamentals of exercise science and can apply this information. In addition, this certification goes beyond the physiological and biomechanical aspects of training. It is our belief that if a client has a question regarding any aspect of the sport, the coach should be knowledgeable in that area. This is why this certification deals with shoe selection, mental training, and much more. This certification was developed in conjunction with a team of expert advisors across many disciplines to ensure that the quality of the certification is second to none.

Like a successful athlete, a successful coach must be confident in their knowledge and the results that they can deliver for the client.

This certification will provide you with the tools necessary to become a knowledgeable, confident, and all-encompassing resource.

Running Certification Structure

The UESCA running certification is not designed to function like a Paint-By-Numbers picture, nor is it designed to offer various coaching templates from which to choose from. As running clients come to coaches with a wide array of goals and issues, the focus of this certification is not to teach exactly how to coach a particular type of client or assign a preformatted program to a client because every client program is different due to the many factors involved. As such, this certification does provide all the requisite theory, information, and tools for you to be able to meet the individual needs of your clients.

Staying current with what is happening within the sport of running and sports science is critical to being an effective coach. Due to the rapid evolution of the sport, many training practices and concepts may not exist in established research. However, by researching the theory behind a training practice, you will likely be able to deduce whether a training practice is valid and applicable to your clients.

With a few exceptions, UESCA rarely takes a definitive standpoint on training practices. Why is this? It is our belief that what might work well for one client might not work so well for another. In other words, people respond differently to the same training stimuli. Therefore it is short sighted to expect that one training methodology adheres to a 'one size fits all' theory. This certification provides you with the science and training theories to be able to advise clients. Therefore, it is your responsibility to decide

what training practice is in your client's best interest.

No Tiers

UESCA is not a tiered coaching certification; meaning, this certification is not a beginner level certification with one or two advanced certification levels above it. As stated earlier, as a certified running coach, your client will expect you to know everything from how to taper for a marathon to what the Rectus Femoris is. At UESCA, we believe a certification program should provide you with all the necessary information to ensure you are a well-educated and up-to-date running coach.

Certification Principles

- One must understand training theory before it can be applied
 - Training must be grounded in science
 - Coaching must take place within your scope of practice
 - Random training = random results
 - Challenge the status quo: do not accept long-standing training practices at face value
 - If you are not constantly learning and applying this knowledge, you are regressing
 - Educated clients are engaged clients
 - Coaching success is largely predicated on thorough training feedback from clients
 - The coaching process is dynamic and not static
 - All coaching decisions must be based on the individual
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Certification Development

What started out as a simple, straightforward process to create this certification became an extremely detailed fact-finding mission to uncover the true science behind endurance sports training, and more specifically, running specific training. What we gleaned from this extensive research was that many of the well-established training principles and "rules" exist solely in folklore or through casual and coincidental relationships.

This is not to say that all current, well-established training practices are incorrect or ineffective, as this is not the case. However, after discovering the alarming amount of incorrect information commonly being applied in endurance sport training practices, the development of this certification program did not take any training theory or practice at face value. Each training methodology described in this certification has been thoroughly vetted to ensure its basis in science and evidence-based practices. We have examined the validity of everything from periodization to hip rotation during the running gait cycle.

If nothing else, the goal of this certification is to challenge you to never accept established norms just because they have been around for a long time. In the words of Nobel Laureate Sir Harold Kroto,

"We need everyone to know how to think rather than accept unquestioningly what we are told. It is amazing to me what people will believe without any proof" (303).

As previously noted, it is important to recognize that accurate training information is a moving target. As new research results become available, current training methodologies are validated, debunked, or modified to reflect the new findings. It is your professional responsibility as well as that of UESCA to stay abreast of the latest training information.

The Sport of Running

Background of Running

Running, as an evolutionary progression, dates back millions of years ago (739). The prevailing theory is that running, and more specifically the ability of humans to run long distances (endurance running) was an evolutionary development that allowed humans to chase animals for food (740). Several physical features of modern day humans such as the Achilles tendon, sweat glands and the gluteus maximus are thought to be the result of this early requirement to run (741).

Competitive running is thought to have originated as part of religious festivals throughout the world (742). For example, the earliest documentation of a competitive running event took place in Ireland in 1829 BC (Tailteann Games) (743). The marathon was first run in the Olympics in 1896 (744).

Running as a training tool was introduced in the 16th century by nobility, likely as a physical conditioning practice for swordsmen. However it wasn't until the 19th century that running was more widely practiced (742).

The popularity of running in the US can be traced back to 1966 when Bill Bowerman (co-founder of Nike®) and W.E. Harris published the book, "Jogging." This book sold millions of copies and is largely responsible for kicking off the running phenomena in the US (745).

Running Events / Distances

As an organized sport, running has several disciplines:

- Road Race (5K–Marathon)

- Track

- Cross Country

Other popular events are trail runs, ultramarathons and obstacle course races (i.e., Tough Mudder®).

The focus of this certification is on road racing and more specifically, the most popular participatory events (5K, 10K, Half Marathon, Full Marathon). While off-road events (trail runs/cross country) and ultra endurance events (ultramarathon) are discussed, the focus is on standard road racing distance events.

Below are the distances of the most popular road racing events:

While these events represent the most popular and common distances, road races (and trail runs) come in all types of distances. For example, an ultramarathon is any type of race that is longer than marathon distance (26.2 miles). These events are held both on and off-road.

If your client is training specifically for a track or ultra-marathon event, it is advised for them to seek out the guidance of a track coach or ultra-marathon coach, respectively.

Participation Categories

Large running races are divided into two main categories: professional/elite and age groups. Age Groupers are amateur athletes and are typically classified by gender and age. This enables amateur athletes to compete against others within their own gender and age range. Some events also have categories for physically challenged athletes.

While not a participation category per se, most medium to large running events start runners based on their running pace (fastest runners in the front, slowest runners in the rear).

Running Statistics

To illustrate the popularity of running, below are statistics from Running USA's "2014 State of the Sport – Part 3: US Race Trends." (747):

- 19,025,000 finishers of US running events
- 42 million people participate in running/jogging
- The number of race finishers has increased 300% since 1990
- Female participation has increased from 25 to 57%
- 28,200 total running events (7% increase from 2012)
- Finishers of road racing events in 2013:
 - 5K: 8.3 million
 - 10K: 1.48 million
 - Half Marathon: 1.96 million
- 61% of participants were women
- Marathon: 541K
- Other: 6.74 million
- The largest age demographic of race finishers was 25–44 yrs (53%)
- Average age: 36.5

From the 2013 National Runner Survey by Running USA (746), core runner demographics:

- 75.8% hold a college degree
- 73.3% have a household income greater than \$75K
- 77.3% participate to stay in shape
- 61.9% participate to relieve stress
- 61.7% participate to have fun

The average competitive runner is middle-aged, affluent, and is looking to stay in shape and wants a challenge. This is the ideal client!

Due to the increasing popularity of running, there is a huge opportunity for certified running coaches.

Certification Content

The content of this certification is derived from our team of expert advisors and professional, peer-reviewed research. In instances where there are small study sizes, we seek out additional, related

studies that may be able to serve as a resource, akin to a meta-analysis (findings based on multiple studies).

While the content of this certification is very robust and unparalleled in regard its scope and detail, it is important to understand that it is by no means an exhaustive source of everything that pertains to running performance.

Our team has identified the most critical areas that pertain to training for running events and more importantly, the areas that are most important to coaches.

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